

Deterministic Zero-Variance Inference & Extreme Value Theory (EVT) Calibration

1. Deterministic OT Midpoint ($t = 0.5$)

$\mathbf{z}_0 = 0 \rightarrow$ Zero-Variance Ray
 $\mathbf{z}_{0.5} = 0.5 \mathbf{z}_{\text{tgt}}$ at $t = 0.5$

Velocity Evaluation:

$$\hat{\mathbf{v}}_{\text{mid}} = v_{\psi}(0.5 \mathbf{z}_{\text{tgt}}, 0.5, \mathbf{z}_{\text{ctx}})$$

$\Rightarrow$ **Zero Monte Carlo noise**

2. Dynamic Velocity Residual

Target Velocity Field:
 $\mathbf{u}_{\text{true}} = \mathbf{z}_{\text{tgt}} - 0 = \mathbf{z}_{\text{tgt}}$

Discrepancy Error Vector:
 $\mathbf{e} = \hat{\mathbf{v}}_{\text{mid}} - \mathbf{z}_{\text{tgt}} \in \mathbb{R}^D$

$\Rightarrow$ Physical dynamic deviation

3. Covariance Whitening (D_M)

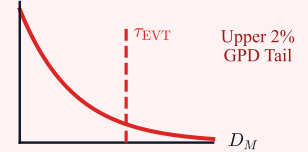
Precision Matrix Calibration:
 $\Sigma^{-1} = (\text{Cov}(\mathbf{e}_{\text{cal}}) + \lambda \mathbf{I}_D)^{-1}$

Whitened Mahalanobis Metric:

$$D_M = \sqrt{\mathbf{e}^T \Sigma^{-1} \mathbf{e}}$$

$\Rightarrow$ Mode-variance scaled

4. EVT Tail Decision



$$\tau_{\text{EVT}} = u + \frac{\sigma}{\xi} \left[\left(\frac{N_u}{N} q \right)^{-\xi} - 1 \right]$$

Decision: $\hat{y} = \mathbb{I}(D_M \geq \tau_{\text{EVT}})$
(Zero Label-Leakage Calibrated)